BIOS13 Question 01 Final Exam
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a) GAs are particularly useful for tackling complex problems (ex. Non-linear, with multiple unknowns, unable to be differentiated). GAs are useful for large problems in which massive combinatorial spaces need to be surveyed. Because of their ability to look at many combinations - they are less likely to get stuck in incorrect local optima. Additionally, GAs can solve either binary or continuous problems, with continuous algorithms able to use real numbers.

The above optimization problem is suitable for a GA. This is largely due to its complex non-linear nature, with the "2*sin(3x)" sinusoidal oscillations making the optimization difficult to solve by hand. These create many local minima. The given domain also makes the problem ideal for a GA to solve, as the combinatorial search space becomes smaller and the GA can be more robust (more likely to reach the global optimum for the solution needed).

b) Biologically, selection describes how individuals with a higher fitness are more likely to reproduce and pass on their genes. In a GA, the population can be sorted by fitness and reproducing parents will then pair in descending fitness order, as mirrors biology (where higher fitness individuals will more likely mate with each other). A GA will also often have a cutoff in which only the top x% of individuals can reproduce, leaving the bottom fitness individuals without offspring. Higher fitness alleles will then naturally be selected for, while lower fitness alleles will be selected against. This models selection pressure of certain gene combinations, and is therefore, a process by which GA can select for global optima.

When reproduction events occur biologically, recombination between the two parents will lead to a new combination of genes. This will create a new chromosome in the population and increase genetic variation. An increase in genetic variation in the population is an opportunity to increase fitness. GAs can model this with a crossover event in which the recombination point is determined at random - providing another opportunity for a GA to find the global optimum.

Additionally, in biological evolution, mutation is important for increasing genetic diversity amongst a population. A mutation can either decrease or increase fitness, or keep it the same. GAs use mutation rates to introduce new alleles and expand the search space for global optima. This adds a random element and an opportunity for higher fitness alleles to appear.

Replacement denotes an entirely new generation. It mirrors biological evolution, as presumably the newer generation has a new order of fitness and new alleles (due to mutation and crossover events). A GA will replace all old offspring with new contenders for a greater likelihood of finding global optima. This is because the new generation offers an opportunity for novel combinations due to unique crossover and mutation events from the previous reproduction session.

c)

```
# Defining a function that takes two parent chromosomes and returns an offspring using
one-point crossover

# example functionality printed to screen using gene_number of 8
gene_number <- 8          # gene number is number of genes in each chromosome (must be
predefined here for my code to work)
```

```r
# function to create two random parental chromosomes to then use in the function for
Question1c
Create_parents <- function(gene_number) {    # This function creates a random set of
two parent chromosomes and takes in gene_number as input
 ch1 <- c()
 ch2 <- c()
  for(gene_index in 1:gene_number) {      # For each gene in the chromosome
   if(runif(1) < 0.5) {                   # fill genes with either 0 or 1
     ch1[gene_index] <- 0            # 50% prob of adding 0 or 1
   } else {
     ch1[gene_index] <- 1           # append to parent 1 chromosome by gene_index
until filled
   }
 }

 for(gene_index in 1:gene_number) {      # restart the for loop to fill parent 2
chromosome in the same way as above
     if(runif(1) < 0.5) {
       ch2[gene_index] <- 0
     } else {
       ch2[gene_index] <- 1
     }
    }

 return(list(ch1 = ch1, ch2 = ch2))  # return a list with parent 1 and parent 2
chromosomes
}

# actual function for question1c
offspring <- function(parents_list) {   # function takes in the parents_list generated
by the function above
 offspring <- c()  # opens and empty vector for the one offspring produced
 crossover <- sample.int(gene_number, 1) # choose a random crossover point along the
chr (in our ex case among 8 genes)
 for(gene_index in 1:gene_number) {   # loop through the gene_index in the total genes
   if(gene_index <= crossover) {     # if the index is less than or equal to the
crossover point --> append genes from parent 1 until the index in the offspring vec
     offspring[gene_index] <- parents_list$ch1[gene_index]
   } else {   # if the index is greater than the crossover point, do the same for the
rest of the gene_indexes but from parent 2
     offspring[gene_index] <- parents_list$ch2[gene_index]
```

```
   }
 }
 return(offspring)  # return the one offspring produced after reproduction and one
crossover event
}

# to show functionality
parents_list <- Create_parents(gene_number)  # create two random parental chromosomes
offspring <- offspring(parents_list)  # perform one reproduction event with one point
of crossover

# printing result to screen using cat for vectors
cat("The offspring of randomly generated parent 1 and 2 is:" , offspring)
```

d)

```
# Defining a function that takes two parent chromosomes and returns an offspring using
one-point crossover and introduces a 0.1 mutation rate

gene_number <- 8         # gene number is number of genes in each chromosome (must be
predefined here for my code to work)
mutation_rate <- 0.1     # defining the mutation rate

# function to create two random parental chromosomes to then use in the function for
Question1c
Create_parents <- function(gene_number) {    # This function creates a random set of
two parent chromosomes and takes in gene_number as input
 ch1 <- c()
 ch2 <- c()
  for(gene_index in 1:gene_number) {      # For each gene in the chromosome
   if(runif(1) < 0.5) {                   # fill genes with either 0 or 1
     ch1[gene_index] <- 0            # 50% prob of adding 0 or 1
   } else {
     ch1[gene_index] <- 1           # append to parent 1 chromosome by gene_index
until filled
   }
 }

 for(gene_index in 1:gene_number) {      # restart the for loop to fill parent 2
chromosome in the same way as above
     if(runif(1) < 0.5) {
       ch2[gene_index] <- 0
     } else {
```

```
      ch2[gene_index] <- 1
     }
    }

 return(list(ch1 = ch1, ch2 = ch2))  # return a list with parent 1 and parent 2
chromosomes
}

# actual function for question1c with modified mutation rate added
offspring <- function(parents_list, mutation_rate) {   # function takes in the
parents_list generated by the function above
 offspring <- c()  # opens and empty vector for the one offspring produced
 crossover <- sample.int(gene_number, 1) # choose a random crossover point along the
chr (in our ex case among 8 genes)
 for(gene_index in 1:gene_number) {   # loop through the gene_index in the total genes
   if(gene_index <= crossover) {     # if the index is less than or equal to the
crossover point --> append genes from parent 1 until the index in the offspring vec
     offspring[gene_index] <- parents_list$ch1[gene_index]
   } else {   # if the index is greater than the crossover point, do the same for the
rest of the gene_indexes but from parent 2
     offspring[gene_index] <- parents_list$ch2[gene_index]
   }
 }

 for(gene_index in 1:gene_number) {     # mutation handled after cross-over event,
looping again through gene index within gene_number
   if(runif(1) <= mutation_rate) {    # generate a random number between 0-1, if the
number is less than or euqal to the mutation rate
     if(offspring[gene_index] == 0) {  #  mutate (so flip from 0 to 1 or 1 to 0 to
model a mutation)
       offspring[gene_index] <- 1
     } else {
       offspring[gene_index] <- 0
     }
   }     # if mutation rate is not satisfied then the function will keep the offspring
bit the same
 }

 return(offspring)  # return the one offspring produced after reproduction and one
crossover event
}
```

```
# to show functionality
parents_list <- Create_parents(gene_number)  # create two random parental chromosomes
offspring <- offspring(parents_list, mutation_rate)  # perform one reproduction event
with one point of crossover

# printing result to screen using cat for vectors
cat("The offspring of randomly generated parent 1 and 2 is:" , offspring)
```

e)

```
# Full GA simulation using population size n=50, for at least 50 generations,
# with ranking selection, recombination, mutation, and a plot of mean and best fitness
over time

install.packages("ggplot2")
library(ggplot2)

##################################################################################
####

gene_number <- 8         # gene number is number of genes in each chromosome (must be
predefined here for my code to work)
pop_size <- 50           # setting pop size
max_generation <- 50     # setting maximum generations
num_breed <- 30          # setting up ranking selection parameter (only 30 best are
allowed to breed)


mutation_rate <- 0.1     # defining the mutation rate, not including any to be saved
from mutation

##################################################################################
####

# instead of just generating two random parents --> generate a random population
Create_pop <- function(pop_size,gene_number) {    # This function creates a random set
of two parent chromosomes and takes in gene_number as input

 genes <- matrix(nrow=pop_size, ncol=gene_number)
  for(ind in 1:pop_size) {               # For each individual:
   for(gene_index in 1:gene_number) {       # For each gene in the chromosome
     if(runif(1) < 0.5) {                    # fill genes with either 0 or 1 (binary
bits)
```

```
      genes[ind,gene_index] <- 0            # 50% prob of added 0 or 1
    } else {
      genes[ind,gene_index] <- 1
    }
  }
}


fitness <- rep(NA,pop_size)    # creates a fitness vec with NA with length of
pop_size (so in our case 50) (will be used in ranking selection)
# return a list with necessary variables for our GA
pop <- list(size=pop_size, gene_number=gene_number, genes=genes, fitness=fitness)
return(pop)
}

##########################################################################################

# Function to calculate fitness for each individual: converting from base 2 (0 1 1 0)
to base 10 (decimal) (for ranking selection)
Evaluate_fitness <- function(pop) {
for(ind in 1:pop$size) {                      # for each index in pop size (1, 2,
3,...50)
  # Calculate the binary number corresponding to the chromosome of ones and zeros:
  Binary_Value <- 0;                          # initate an empty binary value
  for(gene_index in gene_number:1) {          # loop over each gene for the chromosome
but start with the last gene (right most bit is least significant)
    if(pop$genes[ind,gene_index] == 1) {      #if the gene is 1 add the power of 2 (in
binary, each position represents a power of 2 )
      Binary_Value <- Binary_Value + 2^(gene_number - gene_index) # this will be the
current binary value + the gene_number (should be the same for all) minus the index of
the gene starting from the righthand side
    }
  }
  pop$fitness[ind] <- Binary_Value            # store the decimal value for this
individual (all this is doing is filling the fitness parameter of pop)
}
return(pop)
}

##########################################################################################
```

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# Sort the population by fitness for ranked selection:
Sort <- function(pop) {
 # Find order of decreasing fitness:
 new_order <- order(pop$fitness, decreasing=TRUE) #order by the fitness in decreasing
order
 # Next, rearrange the population gene matrix and fitness vector:
 pop$genes <- pop$genes[new_order,]
 pop$fitness <- pop$fitness[new_order]
 return(pop)
}

# best individual is first

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####

# reproduction of entire population --> only allowing top fitness to reproduce
Reproduce <- function(pop,num_breeed) {
 # The population reproduces in pairs, in fitness order
 # First individuals 1 and 2 mate, next individuals 3 and 4, and so on
 # Each pair produces 2 offspring, corresponding to two chromosomes after crossover
 # Offspring replace the parent population from the bottom of the list.
 offspring_pos <- pop$size # where to put the offspring, starting from bottom
 for(parent1 in seq(1,num_breeed,by=2)) {
   parent2 <- parent1 + 1  #loop through the top num_breed individuals in steps of 2
   # crossover event
   crossover <- sample.int(gene_number,1)  # pick a random crossover point along the
chrom
   for(gene_index in 1:pop$gene_number) {  # for each gene in the ch if the gene ind
is before or at the crossover - copy from parent 1 --> offspring 1 and parent 2 -->
offspring 2
     if(gene_index <= crossover) {
       pop$genes[offspring_pos,gene_index] <- pop$genes[parent1,gene_index]  # for
offspring 1
       pop$genes[offspring_pos-1,gene_index] <- pop$genes[parent2,gene_index] # for
offspring 2
     } else { # after the crossover point swap the parents (parent 2 -> offspring 1
and parent 1 -> offspring2)
       pop$Genes[offspring_pos,gene_index] <- pop$Genes[parent2,gene_index]
       pop$Genes[offspring_pos-1,gene_index] <- pop$Genes[parent1,gene_index]
     }  # single point crossover
```

```
   }
   offspring_pos <- offspring_pos - 2 # move two positionns up to place the next pair
of offspring
 }
 return(pop)

 # this does replacement - so if you set the num_breed boundary to 10 - the first 5
pairs will breed and replace the last 10 individuals in the population (the 30 middle
will remain the same before sorting)
}

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####

# mutate the entire population (does not allow for exempt individuals in my model)
Mutate <- function(pop, mutation_rate) {
 for(ind in 1:pop$size) {                           # loop through all ind
   for(gene_index in 1 : pop$gene_number) {         # loop through each gene in the
current ch
     if(runif(1) <= mutation_rate) {                # generate a random number between
0-1, if the number is less than or euqal to the mutation rate
       if(pop$genes[ind,gene_index] == 0) {         #  mutate (so flip from 0 to 1 or
1 to 0 to model a mutation)
         pop$genes[ind,gene_index] <- 1
       } else {
         pop$genes[ind,gene_index] <- 0
       }
     }
   }
 }
 return(pop)
}

##################################################################################
####

population <- Create_pop(pop_size,gene_number)  # call function to create the
population

population <- Evaluate_fitness(population)   # calculate fitness for all individuals
and update population matrix
```

```
population <- Sort(population)        # sort the population matrix chromosomes
according to fitness

##################################################################################
####

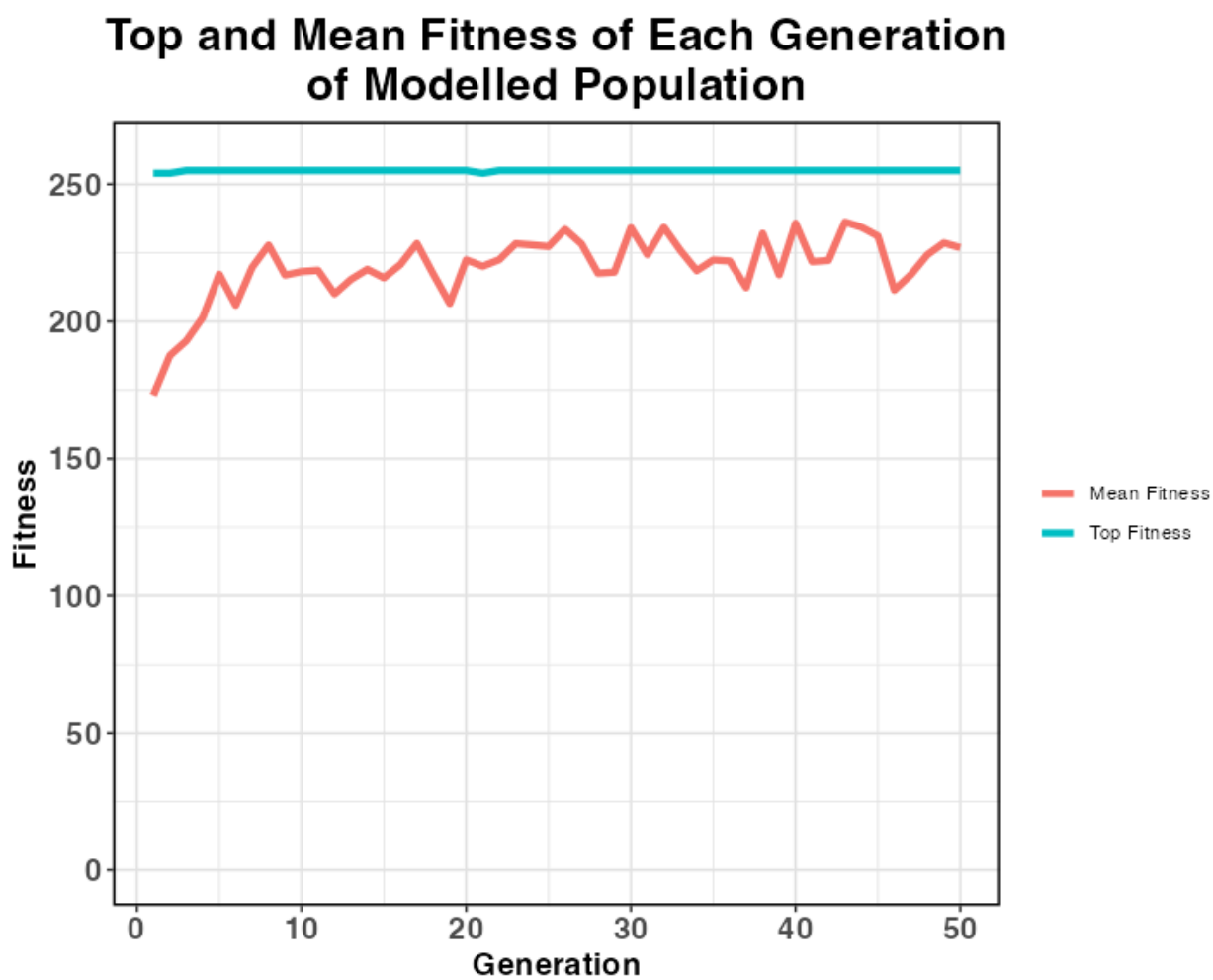
topfitness <- c(0,max_generation) # empty vectors to store results
meanfitness <- c(0,max_generation)

for(generation in 1:max_generation) {
 population <- Reproduce(population,num_breed)
 population <- Mutate(population,mutation_rate)
 population <- Evaluate_fitness(population)   # give back altered population with
fitness
 population <- Sort(population)
  # save statistics:
 topfitness[generation] <- population$fitness[1]
 meanfitness[generation] <- mean(population$fitness)
}


plot_fitness <- ggplot(
 data = data.frame(x = seq(1,max_generation, by=1), topfitness = topfitness,
meanfitness = meanfitness),
 aes(x = x)) +
 geom_line(
   aes(y = topfitness, color = "Top Fitness"),
   linewidth = 1.5) +
 geom_line(
   aes(y = meanfitness, color = "Mean Fitness"),
   linewidth = 1.5) +
 scale_y_continuous(limits = c(260,0), # y - axis limits
                    breaks = seq(0,260, by = 50)) + # adjusting y axis spacing and tick
marks
 labs(   # adding labels
 title = ("Top and Mean Fitness of Each Generation\nof Modelled Population"),
 y = ("Fitness"),
 x = ("Generation"),
 color = NULL
 ) +
 theme( # adjusting aesthetics
   panel.background = element_rect(fill = "white"),
```

```
  panel.grid.major = element_line(color = "grey90"),
  panel.grid.minor = element_line(color = "grey90"),
  panel.border = element_rect(color = "black", fill = NA, linewidth = 1),
  legend.position = "right",
  plot.title = element_text(size = 20, face = "bold", hjust = 0.5),
  axis.title.x = element_text(size = 14, face = "bold"),
  axis.title.y = element_text(size = 14, face = "bold"),
  axis.text.x = element_text(size = 14, face = "bold"),
  axis.text.y = element_text(size = 14, face = "bold"))


print(plot_fitness)  # printing plot of best fitness of each generation
```



f) This may limit the GA's ability to find the global optimum of the function as the combinatory search space is limited to the initial population as parental chromosomes. While recombining will create new combinations over generations, these have a limit. Mutation can introduce new solutions in the

population, but with a relatively low mutation rate, this can take many iterations before significantly impacting the allelic frequency. This is a pitfall of GAs and can lead to the simulations getting stuck in local optima.